

Solution Requirements

Date	6 July 2026
TeamID	
Project Name	Plugging into the Future:An Exploration of electricity consumption patterns
Maximum Marks	4Marks

Define Functional and Non-Functional Requirements:

Define the requirements of the **Electricity Consumption Prediction System** to specify what the system should do and how well it should perform.

- **Functional Requirements(FR):** Describe the features of the system,such as collecting electricity consumption data ,predicting future electricity consumption, and displaying the prediction results
- **Non-Functional Requirements(NFR):**Describe the quality of the system,including performance, security, reliability, and usability.

Fill out the descriptions and necessary details for each category provided below.

Step1:Functional Requirements(FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using a username and password before accessing the prediction system	High
2	Authorization levels	Admin can manage the dataset and prediction model. Users can enter electricity consumption data and view prediction results.	High

3	External interfaces	The system inter acts with the Flask web application, machine learning model,and electricity consumption database	High
4	Data processing	The system accepts electricity consumption data, preprocesses the data, performs prediction, and displays the predicted electricity consumption.	High
5	Reporting	The system displays prediction results and model accuracy reports.	Medium
6	Businessrules, etc.	Prediction is based on historical electricity consumption, season, weather conditions, time, and other input features	High
7	Compliance to laws or regulations	The system protects user data and follows applicable data privacy guidelines.	High
8	Other	The system validates user input before making predictions.	Medium

Step2:Non-Functional Requirements(NFR)

S.No	NFRCategory	Requirement Description	Target Metric / AcceptanceCriteria
1	Performance& Speed	The system should generate prediction results quickly.	Prediction displayed within 2 seconds)
2	Scalability	The system should support multiple users and larger datasets.	Supports 100+ concurrentusers
3	Security&Data Privacy	Applicant information should be securely stored and protected.	Password encryption and secure data storage
4	Reliability& Availability	The application should be available when ever users need it.	99%uptime
5	Usability &Accessibility	The web interface should be simple,responsive,and easy to use.	User-friendly inter face with responsive design

6	Other	The system should be easy to maintain and update with new machine learning models.	Modular and maintainable code
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